

Google's Self-Driving Vehicle

Second Generation, 2012

Google's self-driving vehicles understand where they are and what's around them through sensors that are purpose-built to help the vehicles perceive their surroundings accurately, and software that processes the information received.

Laser

This sensor gives the vehicle a 360-degree understanding of its environment so the car can sense objects in front of, beside, and behind itself at the same time, all the time. The laser also helps the vehicle to determine its location in the world.

Processor

Information from the sensors is cross-checked and processed by the software so that different objects around the vehicle can be sensed and differentiated accurately, and safe driving decisions can then be made based on all the information received.

Position sensor

This sensor, located in the wheel hub, detects the rotations made by the wheels of the car to help the vehicle understand its position in the world.

Orientation sensor

Similar to the way a person's inner ear gives them a sense of motion and balance, this sensor, located in the interior of the car, works to give the car a clear sense of orientation.

Radar

This sensor detects vehicles far ahead and measures their speed so that the car can safely slow down or speed up with other vehicles on the road.



Safety drivers

Drivers also test the vehicles daily, reporting feedback on how to make the ride more safe and comfortable.



A look at the second generation of Google's self-driving vehicle

Tesla, BMW and Google are the current front-runners when it comes to development of driverless cars. And Tesla's Elon Musk was in talks with Google to bring improved self-driving cars for the future. Tesla's very own Tesla Autopilot system in the Model S car is touted to be one of the most advanced and incremental driving system of the current generation. Couple that with the use of powerful cars, high quality sensors and innovative software and what you have is a lot of potential when it comes to making driverless cars a reality which is actually practical. Even BMW's demo of its modified electric car – the BMW i3 at CES 2015 which could drive and park itself was really interesting. What's more amazing is that, you can call this car right next to where you are on the footpath by simply making use of a smartwatch app. Watch a video of it in action here: <http://dgit.in/BMWi3AutoPark>

The transition to autonomous cars:

Well, we've seen the amount of promise that driverless car technology holds especially in terms of the Internet of Things and then comes the next question: How will the transition to self-driving cars actually work? Well, we have an answer. The transition to self-driving cars can take place in three stages: